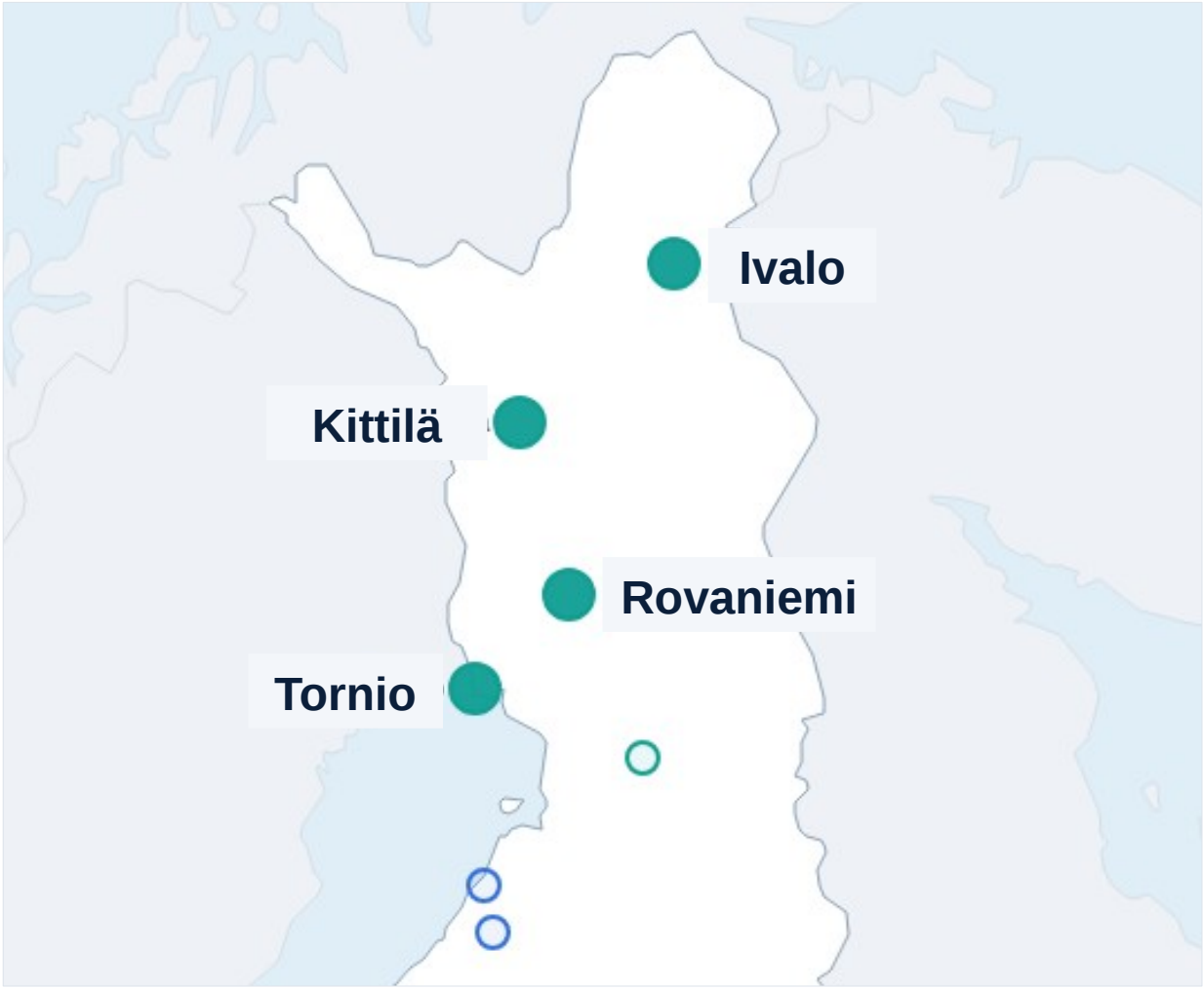
In May 2005, four Lapland rivers hit flood levels together.

Lapland's snow had warned us two months earlier.

Flow over 1-in-10 level · snow top 15 % · in-sample



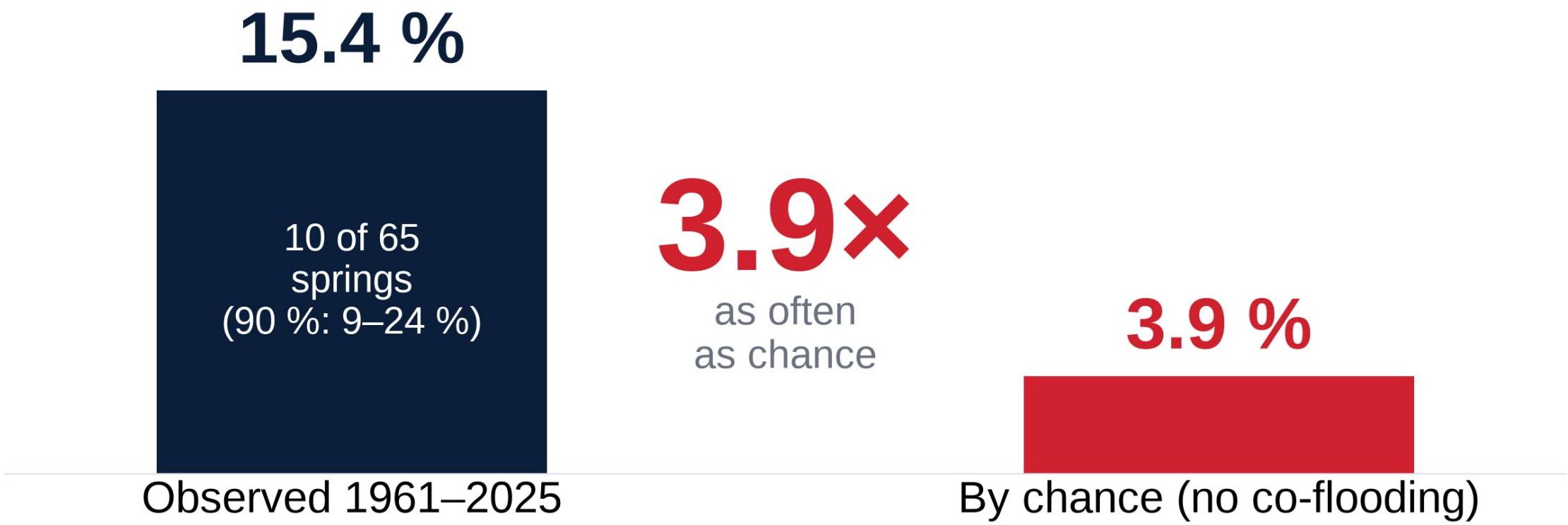
Tulva app, spring 2005: filled = over its 1-in-10 flood level

Source: SYKE discharge 1961–2025 and 1 March snow (SWE)

THE GAP

Finland's rivers flood together, not one at a time

Springs when ≥4 of 14 river areas pass their 1-in-10 flood level



Cover is bought in autumn. The snow signal comes in March.

Source: SYKE 1961–2025 (+ SMHI Torne); chance = rivers flood independently, same gauge-years as observed

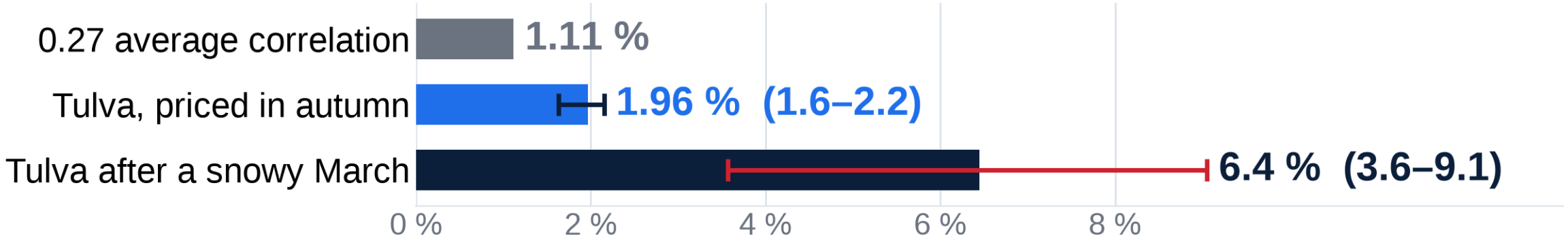
THE PRODUCT

Tulva Index: sold in autumn, updated in March

Pays if ≥2 of 5 northern rivers pass their 1-in-50 flood level

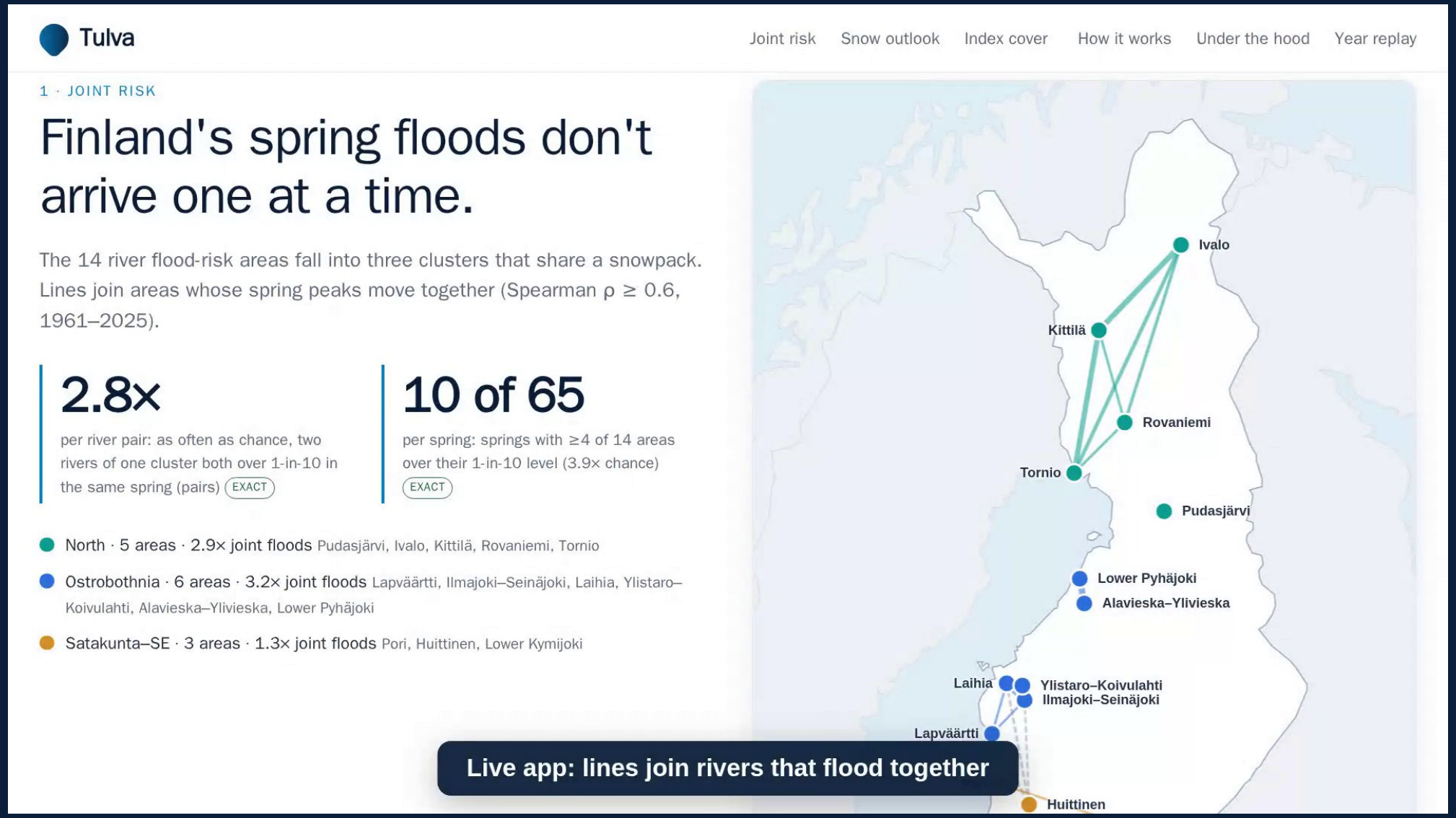
Fair price ≈1.8× a model using the 0.27 average correlation of the 2027 Finnish flood-zone matrix

EXPECTED LOSS, % OF THE LIMIT PER YEAR (90 % INTERVALS)



After a snowy March the risk triples (1.96 % → 6.4 %; 90 %: 2.1–4.4×)
Priced in autumn. In March the insurer acts on it: top-up cover, claims staff in the regions at risk.

Sources: SYKE 1961–2025, Tulva model, year-bootstrap intervals; 0.27 = mean of the 19-zone matrix, Del. Reg. (EU) 2026/269

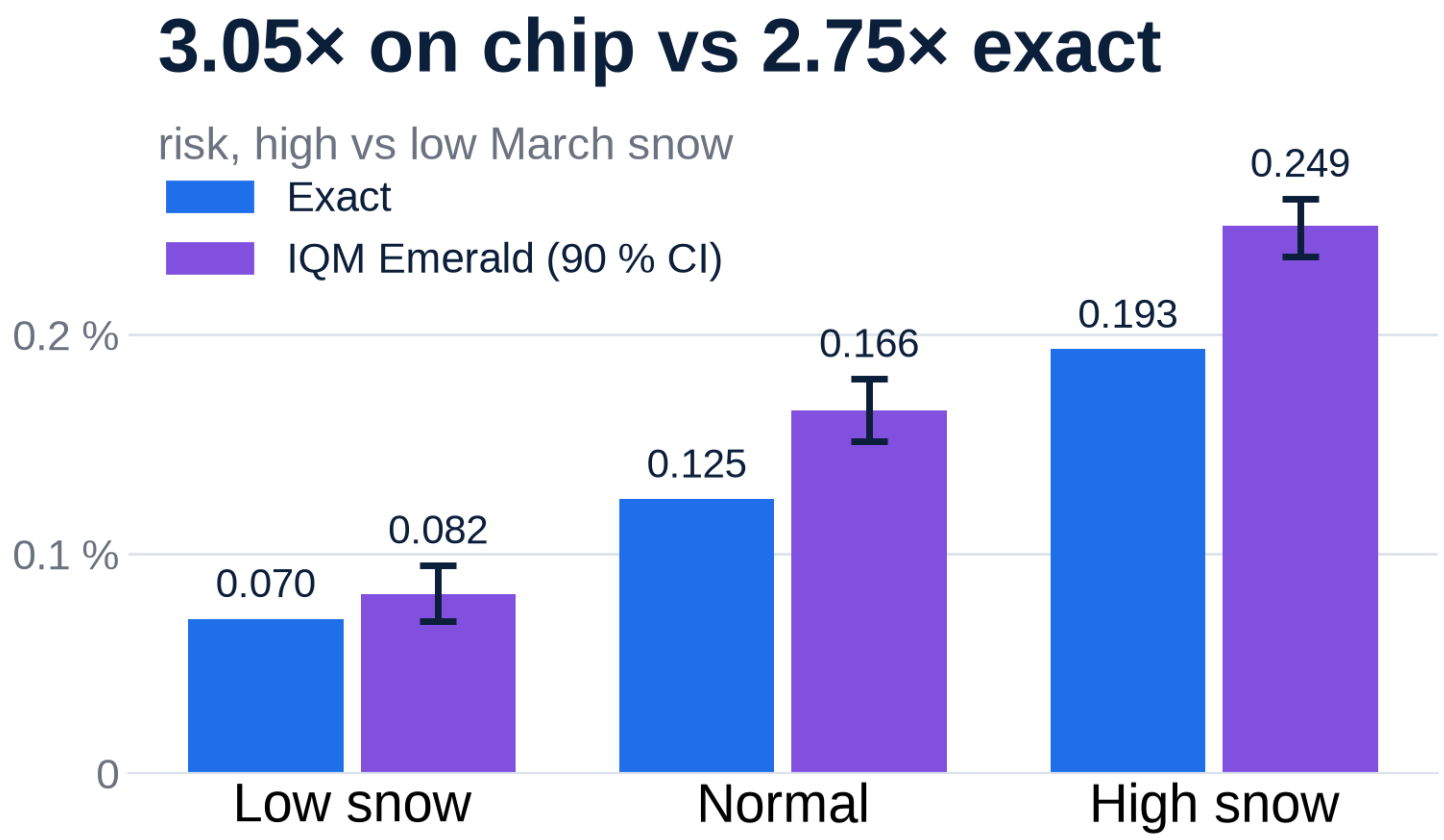


Live app: joint-flood map → March snow → quantum chip → 2005 replay

QUANTUM, IN PRACTICE

It runs on four real quantum computers today

Two-river North-cluster pair on IQM Emerald



CALIBRATED SAMPLING + SHALLOW AMPLITUDE ESTIMATION

Two-river core

IBM Marrakesh, IQM Garnet and Emerald: 3 qubits, Grover steps k = 0–2

14-river model

IQM Emerald: 18 qubits, sampling only

Amplification

amplified circuits keep ≈0.5 of the signal on Emerald, 0.6–0.8 on IBM / Garnet

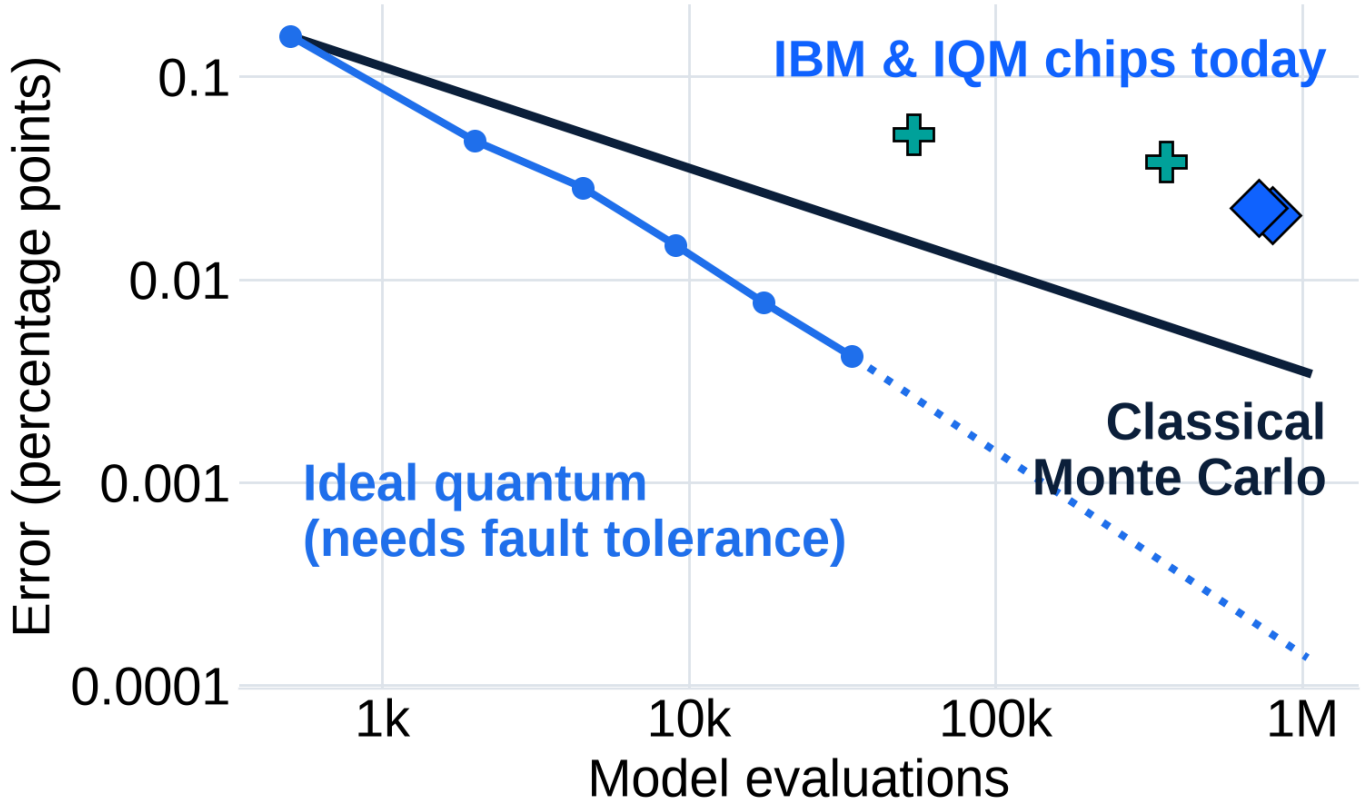
VTT Q50, Finland's own

3 repeats: 0.11–0.53 % vs 0.125 % exact. Calibration not yet stable

Intervals, raw readings and job IDs: appendix

QUANTUM, HONESTLY

Classical Monte Carlo still wins today



Two-river North-cluster pair, both over 1-in-50: error vs effort

Method ready, hardware not yet

Today: rare floods drown in noise.

14 rivers, 18 qubits:
≥4 over 1-in-10: exact 10.9 %; IQM Emerald 13.8 %, VTT Q50 31 %
Emerald: correlation attenuated (1.4× vs exact 2.5×); 1-in-50: 16 % vs 4.8 % exact

Prices are computed classically; quantum is the R&D track.

BUSINESS

Who pays whom, and why now

BUYER

Finnish home insurers

e.g. If Insurance

→

RISK CARRIER

A reinsurer

Premium, insurer → reinsurer: ≈€0.5–0.7M a year per €10M layer

→

TULVA

Index + pricing licence

Licence → Tulva: €25k pilot, then ~€75k a year

→

YEAR 3

3–6 insurers × €75k ≈ €0.2–0.45M/yr

Upside: Nordic snowmelt (assumption)

Why now **From 2027 standard-formula insurers get fixed flood correlations that ignore the snow; internal-model insurers like If Insurance get a snow-aware joint-flood number for their own model.**

Basis risk **same 1-in-50 idea Finnish home policies use, but on river flow. Back-test: paid in the 1966, 2005 and 2018 flood springs; missed Ostrobothnia 2012–13 and 2024 (snow flag on: a false alarm).**

Ask **One insurer's flood claims for 8 weeks → basis-risk map + firm quote for 1 Jan 2028.**

Del. Reg. (EU) 2026/269 · back-test at 1-in-10 (at 1-in-50: 1966, 2005) · premium = model × 2.5–3.5 · prices: assumptions

Tulva

Next March, the snow will warn us again.

This time, it will have a price.

BUILT THIS WEEKEND

SYKE data pipeline | Joint flood model | Index pricing | Live app | QAE on IBM, IQM & VTT Q50

RESOURCES USED

SYKE open data | IBM Marrakesh | IQM Garnet & Emerald (Resonance) | VTT Q50 (LUMI) | Qrisp | Qiskit | AI coding agents

Toivo Kallio · **Vaishnav Rane**

Team 21 STUHI · Quantum x Finance 2026, Hanken × Ultrahack

APPENDIX

Hardware runs in detail

Chip	Circuit	Exact	Raw	Calibrated (90 % CI)	Job
IBM Marrakesh	North-cluster pair, 3 qubits, k = 0–2	0.125 %	0.262 %	**0.149 % (0.140–0.157)**	db3tddsvf2bc73ctoprg
IQM Garnet	North-cluster pair, 3 qubits, k = 0–2	0.125 %	0.368 %	**0.164 % (0.147–0.180)**	01a11cc3-7f6c…
IQM Emerald	Snowy / low March ratio, same pair, k = 0–2	2.75×	1.3×	**3.05× (2.6–3.6)**	product-…075338Z
IQM Emerald	14 rivers ≥4 over 1-in-10, 18 q / 44 CZ	10.9 %	–	**13.8 % (readout-corr.)**	01a11cd4-da38…
IQM Emerald	14 rivers ≥2 over 1-in-50, same circuit	4.8 %	–	**16.0 % (fails)**	01a11f9b-c62a…
IQM Emerald	Control: independent circuit, ≥4 over 1-in-10	4.4 %	–	**9.7 % (noise fattens tails)**	…T135910Z
VTT Q50	Same pair, 3 repeats: not stable	0.125 %	1.48–1.58 %	**0.47 / 0.53 / 0.11 %**	7d20d8ac / a6fb9d52 / 381179fb
VTT Q50	14 rivers ≥4 over 1-in-10, 18 q / 49 CZ	10.9 %	34.2 %	**31.2 % (readout-corr.)**	9cdb0158-7574…

Calibrated = noise-aware MLAE, two known-answer calibration circuits in the same job; bootstrap 90 % intervals. IBM / Garnet intervals exclude exact (+19 % / +31 % bias). Pair = two rivers with identical North-cluster loadings and 1-in-50 trigger. Classical Monte Carlo at the same budget is more precise (±0.004 pp).

Signal kept by amplification (k = 1 / 2): Emerald 0.56–0.59 / 0.44; VTT Q50 0.12–0.19 / 0.05–0.06. Q50 zero-calibration offset moved 0.002 → 0.006 between repeats, driving the 5× spread.